

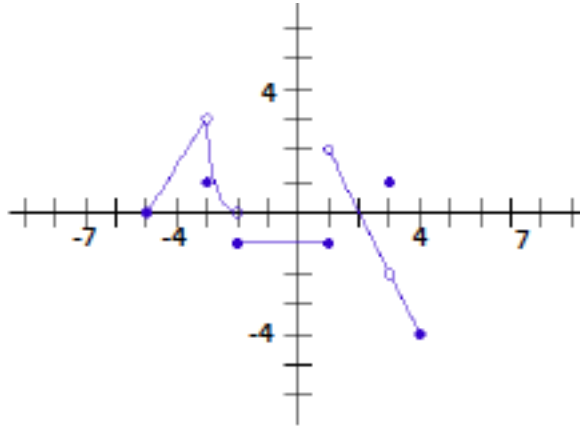
Your Name:

ID #:

Worksheet: Limits at Continuity (Discontinuity)-Part1

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Refer to the graph below, state with explanations whether the following limits exists.



Determine if the following limits exists:

$$\lim_{x \rightarrow -3} f(x)$$

$$\lim_{x \rightarrow -2} f(x)$$

$$\lim_{x \rightarrow 0} f(x)$$

$$\lim_{x \rightarrow 1} f(x)$$

$$\lim_{x \rightarrow 2} f(x)$$

$$\lim_{x \rightarrow 3} f(x)$$

Solution:

1. $\lim_{x \rightarrow -3} f(x) = 1$ since $\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^-} f(x) = 1$

2. $\lim_{x \rightarrow -2} f(x) = DNE$

$$\lim_{x \rightarrow -2^+} f(x) = -1 \text{ but } \lim_{x \rightarrow -2^-} f(x) = 0$$

$$\lim_{x \rightarrow -2^+} f(x) \neq \lim_{x \rightarrow -2^-} f(x) = 0$$

3. $\lim_{x \rightarrow 0} f(x) = -1$ since $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = -1$

4. $\lim_{x \rightarrow 1} f(x)$: DNE (right and left limits differ).

$$\lim_{x \rightarrow 1^+} f(x) = 2 \text{ but } \lim_{x \rightarrow 1^-} f(x) = -1$$

$$\lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x) = 0$$

5. $\lim_{x \rightarrow 2} f(x) = 0$ since $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x) = 0$

6. $\lim_{x \rightarrow 3} f(x) = -2$ since $\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^-} f(x) = -2$

2. (a) $\lim_{x \rightarrow c} (2x + 5) = 2C + 5$

(i) $\lim_{x \rightarrow 0} e^x = e^0 = 1$

(b) $\lim_{t \rightarrow 6} 8(t - 5)(t - 7) = 8(6 - 5)(6 - 7)$
 $= -8$

(j) $\lim_{x \rightarrow -5} \frac{x^2 + 3x - 5}{x + 7} = \frac{25 - 15 - 5}{-5 + 7} = \frac{5}{2}$

(c) $\lim_{x \rightarrow 2} \frac{x + 2}{x^2 - 5x + 6}$
 $= \frac{\lim_{x \rightarrow 2} (x + 2)}{\lim_{x \rightarrow 2} (x^2 - 5x + 6)} = \infty$

(k) $\lim_{x \rightarrow 2} \frac{e^x}{x + 1} = \frac{e^2}{3}$

(l) $\lim_{x \rightarrow 3} \frac{x^2 - 3}{\ln e^x} = \frac{3^2 - 3}{\ln e^3} = \frac{6}{3} = 2$

(d) $\lim_{x \rightarrow 0} \frac{x^3 + (x + 1)\ln(x + 3)}{x^3 - 4}$
 $\stackrel{\text{plug in 0}}{=} \frac{\ln 3}{-4}$

(m) $\lim_{x \rightarrow 0} \sqrt[3]{x^4 + 2x^3 + 8} = \sqrt[3]{0^4 + 2 \cdot 0^3 + 8} = 2$

(e) $\lim_{x \rightarrow 4} x^{x+1} = 4^{4+1} = 4^5$

(n) $\lim_{x \rightarrow 0} (x^2 + 1)^{2x-2} = (0^2 + 1)^{2 \cdot 0 - 2} = 1$

(f) $\lim_{x \rightarrow -2} e^{x+1} = e^{-2+1} = \frac{1}{e}$

(o) $\lim_{x \rightarrow -1} e^{x^2} = e^{(-1)^2} = e^1 = e$

(g) $\lim_{x \rightarrow \frac{\pi}{4}} \sin(2x) = \sin(2 \cdot \frac{\pi}{4})$
 $= \sin(\frac{\pi}{2}) = 1$

(p) $\lim_{x \rightarrow \frac{\pi}{6}} \frac{\tan(x)}{\cos(2x)} = \frac{\tan(\frac{\pi}{6})}{\cos(2\frac{\pi}{6})} = \frac{\frac{1}{\sqrt{3}}}{\frac{1}{2}} = \frac{2}{\sqrt{3}}$

(h) $\lim_{x \rightarrow \frac{\pi}{2}} \csc(x) = \csc(\frac{\pi}{2}) = \frac{1}{\sin(\frac{\pi}{2})} = 1$

3. Show all the reasons to find the following limits:

(a) $\lim_{x \rightarrow 0+} e^{\frac{1}{x}} =$

(b) $\lim_{x \rightarrow 0-} e^{\frac{1}{x}} =$

Solution:
 $\lim_{x \rightarrow 0+} \frac{1}{x} = \infty$
 $\Rightarrow \lim_{x \rightarrow 0+} e^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0+} (\frac{1}{x})} = e^{\infty} = \infty$

Solution:
 $\lim_{x \rightarrow 0-} \frac{1}{x} = -\infty \Rightarrow \lim_{x \rightarrow 0-} e^{\frac{1}{x}} =$
 $\lim_{x \rightarrow 0-} (\frac{1}{x}) = e^{-\infty} = \frac{1}{e^{\infty}} = 0$

(c) $\lim_{x \rightarrow 0^+} \ln x =$

(d) $\lim_{x \rightarrow 0^-} x e^{\frac{1}{x}} =$

Solution:

Refer to the graph of $\ln x$, as $x \rightarrow 0^+$,

$\ln x \rightarrow -\infty$

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

Solution:

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$\Rightarrow \lim_{x \rightarrow 0^-} (e^{\frac{1}{x}}) = e^{-\infty} = \frac{1}{e^{\infty}} = 0 \Rightarrow$$

$$\lim_{x \rightarrow 0^-} x e^{\frac{1}{x}} = \lim_{x \rightarrow 0^-} x \cdot \lim_{x \rightarrow 0^-} (e^{\frac{1}{x}}) = 0 \cdot 0 = 0$$

4. Show all the work to find the following limits:

(a) $\lim_{x \rightarrow 0} \frac{\sin(3x)}{4x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{4x} \stackrel{\text{L.H}}{=} \lim_{x \rightarrow 0} \frac{(\sin(3x))'}{(4x)'} = \lim_{x \rightarrow 0} \frac{3 \cos(3x)}{4} = \frac{3}{4}$$

(b) $\lim_{x \rightarrow 1} \frac{5x^4 - 4x^2 - 1}{10 - x - 9x^3}$

Solution:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{5x^4 - 4x^2 - 1}{10 - x - 9x^3} &\stackrel{\text{L.H}}{=} \lim_{x \rightarrow 1} \frac{(5x^4 - 4x^2 - 1)'}{(10 - x - 9x^3)'} \\ &= \lim_{x \rightarrow 1} \frac{20x^3 - 8x}{-1 - 27x^2} = \frac{20 - 8}{-1 - 27} = -\frac{3}{7} \end{aligned}$$

(c) $\lim_{x \rightarrow -1} \frac{\sqrt{x+4} - 3}{x+1} =$

Solution:

$$\lim_{x \rightarrow -1^+} \frac{\sqrt{x+4} - 3}{x+1} = \frac{\sqrt{-1+4} - 3}{-1^+ + 1} = \frac{\sqrt{3} - 3}{0^+} = \infty,$$

$$\lim_{x \rightarrow -1^-} \frac{\sqrt{x+4} - 3}{x+1} = \frac{\sqrt{-1+4} - 3}{-1^- + 1} = \frac{\sqrt{3} - 3}{0^-} = -\infty,$$

left side limit is not equal to the right side limit, then $\lim_{x \rightarrow -1} \frac{\sqrt{x+4} - 3}{x+1} = DNE$.